



SIH26167 - COMPLETE WINNING ANALYSIS

Yash, you've made your decision. Now let's dissect every word of this PS and build the exact winning blueprint.



PART 1: PS DISSECTION - LINE BY LINE

What ISRO is REALLY asking:

text

PS REQUIREMENT BREAKDOWN:

MANDATORY COMPONENT 1: Remote Sensing Adaptation

"At least one visual or vision-language component must be fine-tuned or adapted using BigEarthNet"

Translation:

- You CANNOT use vanilla BLIP-2/LLaVA
 - Must show domain adaptation
 - BigEarthNet = Sentinel-1 SAR + Sentinel-2 Multispectral + Text
 - This is NON-NEGOTIABLE
-

MANDATORY COMPONENT 2: Single Image VQA

"Visual question answering shall be mandatory"

Translation:

- User uploads one satellite image
 - User asks natural language question
 - System answers accurately
 - Must work on VRSBench + RSVQA benchmarks
 - SCORED QUANTITATIVELY
-

MANDATORY COMPONENT 3: Change Detection

"Change description or change-based VQA from bi-temporal image pair is mandatory"

Translation:

- Upload image from 2020 + image from 2024
 - System detects what changed
 - System describes changes in text
 - System generates change map
 - Must work on CDVQA benchmark
 - SCORED QUANTITATIVELY
-

MANDATORY COMPONENT 4: SAR + Optical Fusion

"Extract complementary information from co-registered optical and SAR image pair"

Translation:

- Upload optical image + SAR image
 - Same location, different sensors
 - System combines both for better answer
 - SAR sees through clouds
 - Optical gives color/texture
 - Together = complete picture
-

MANDATORY COMPONENT 5: Agentic Orchestration
"Automatically select, sequence, execute appropriate specialist models per query"

- Translation:
- User asks question
 - Agent decides WHICH model to use
 - Agent sequences multiple models
 - Agent combines outputs
 - Agent explains what it did (audit log)
 - THIS IS YOUR STRONGEST COMPONENT

EVALUATION DATASETS:

- VRSBench: Single image VQA + captioning
- RSVQA: Single image visual QA
- CDVQA: Change detection VQA
- ISRO's OWN: Cartosat-2S + RISAT

(This is the UNKNOWN - cannot prepare)

 PART 2: TECHNICAL ARCHITECTURE REQUIRED

Complete System You Need to Build:
text

SATQUERY AI - COMPLETE ARCHITECTURE

LAYER 1: INPUT PROCESSING

INPUT HANDLER
• GeoTIFF/TIFF parser (rasterio) • PNG/JPEG for benchmarks • SAR image preprocessor • Optical image preprocessor • Image pair validator • Co-registration checker • Metadata extractor (sensor type, date, resolution) • Format compatibility checker

LAYER 2: AGENTIC CONTROLLER (YOUR STRENGTH)

ORCHESTRATION ENGINE
• Query intent classifier → VQA? Captioning? Change? Fusion? • Input configuration detector → Single? Bi-temporal? Cross-modal? • Model registry manager → Which model for which task • Task sequencer → What order to run models • Output combiner → Merge text + spatial results • Audit log generator → What ran, what parameters • Confidence estimator → How confident is the answer

LAYER 3: SPECIALIST MODEL REGISTRY

MODEL REGISTRY
MODEL 1: RS-VQA Engine Purpose: Answer questions about single image Base: Fine-tuned BLIP-2 / GeoChat Training: BigEarthNet + RSVQA dataset Input: Optical/SAR image + question Output: Text answer + confidence score
MODEL 2: RS-Captioning Engine Purpose: Describe satellite image Base: Fine-tuned RemoteCLIP / BLIP Training: BigEarthNet + VRSBench Input: Single image Output: Detailed scene description
MODEL 3: Change Detection Engine Purpose: Detect changes between two images Base: ChangeFormer / BIT / SNUNet Training: LEVIR-CD / SYSU-CD / WHU-CD Input: Image T1 + Image T2 Output: Change map + change description
MODEL 4: Change VQA Engine Purpose: Answer questions about changes Base: Fine-tuned on CDVQA dataset Input: Bi-temporal pair + question Output: Text answer about changes
MODEL 5: SAR-Optical Fusion Engine Purpose: Extract info from both modalities Base: Cross-modal attention / MAREsNet Input: SAR image + Optical image Output: Fused analysis + segmentation map
MODEL 6: Region Grounding Engine Purpose: Locate objects mentioned in query Base: RSVG / GeoRSClip grounding Input: Image + text query Output: Bounding boxes + highlighted region

↓

LAYER 4: OUTPUT GENERATION

OUTPUT LAYER
• Text response generator • Visual overlay creator (bounding boxes, change maps, segmentation overlays) • Confidence visualization • Execution summary generator • Report PDF exporter • Map visualization (Leaflet.js)

↓

LAYER 5: FRONTEND (Web Application)

WEB INTERFACE
• Image upload interface

- Query input box
- Image type selector (Single/Bi-temporal/Cross-modal)
- Real-time agent status display
- Results panel (text + visual)
- Execution audit log panel
- Download report button
- Example queries panel

 PART 3: MODELS YOU NEED - SPECIFIC NAMES
Every model mapped to every requirement:
text

REQUIREMENT → MODEL → STATUS

1. RS-ADAPTED VLM (MANDATORY):

Option A: GeoChat
 → Already RS fine-tuned ✓
 → Handles RS VQA natively
 → Available on HuggingFace
 → Paper: arxiv.org/abs/2311.15826
 → USE THIS AS BASE

Option B: RemoteCLIP
 → RS-adapted CLIP model
 → Good for image-text matching
 → Available on HuggingFace
 → Paper: arxiv.org/abs/2306.11029

Option C: RSGPT
 → RS-specific GPT variant
 → VQA + captioning capable

RECOMMENDATION: GeoChat as primary
 Fine-tune further on BigEarthNet

2. CHANGE DETECTION MODEL:

Option A: ChangeFormer
 → Transformer-based change detection
 → State of art on LEVIR-CD
 → GitHub: [justchenhao/ChangeFormer](https://github.com/justchenhao/ChangeFormer)

Option B: BIT (Binary Change Detection)
 → Bitemporal Image Transformer
 → Fast and accurate
 → GitHub: [justchenhao/BIT_CD](https://github.com/justchenhao/BIT_CD)

Option C: SNUNet
 → Dense skip connections
 → Good for subtle changes

RECOMMENDATION: ChangeFormer
 + Fine-tune on CDVQA for VQA ability

3. SAR PROCESSING:

Option A: SAR-RARP50

→ SAR specific processing

Option B: Custom CNN on Sentinel-1

- Train on BigEarthNet SAR bands
- Extract SAR features

Option C: PreSARNet

- Pre-trained SAR network

KEY INSIGHT:

BigEarthNet has Sentinel-1 SAR data

- = Use it to train SAR feature extractor
- = Then fuse with optical features

Fusion Approach:

- Extract SAR features (CNN)
- Extract Optical features (ViT/ResNet)
- Cross-attention fusion layer
- Joint representation for analysis

4. GROUNDING MODEL:

Option A: RSVG (RS Visual Grounding)

- Specifically for satellite images
- Returns bounding boxes
- Paper: arxiv.org/abs/2210.12634

Option B: GeoRSCLIP + Grounding DINO

- Combine RS CLIP with grounding
- More flexible approach

5. AGENTIC FRAMEWORK:

LangGraph (YOUR EXPERTISE)

- StateGraph for agent workflow
- Tool calling for model invocation
- Memory for multi-turn queries
- Routing logic for model selection

This is WHERE YOU DOMINATE

Nobody else will have 35-agent
experience like your CREVA work



PART 4: DATASETS - COMPLETE BREAKDOWN
text

DATASETS YOU NEED:

1. BigEarthNet (PRIMARY - MANDATORY)

Size: 590,326 image patches
Sensors: Sentinel-1 SAR + Sentinel-2 MS
Labels: 43 land-cover classes
Use: Fine-tune your VLM
Download: bigearth.net
Size on disk: ~65GB

IMPORTANT: The PS references

arxiv.org/abs/2603.29630

This is BigEarthNet-v2 with TEXT
annotations = perfect for VLM training

2. VRSBench (EVALUATION)
Purpose: VQA + Captioning + Grounding
Use: Evaluate your single-image models
Download: Available on HuggingFace
Must score well [HERE](#)
3. RSVQA (EVALUATION)
Purpose: Remote sensing VQA
Two versions: LR (low res) + HR (high res)
Use: Evaluate VQA capability
Download: rsvqa.reactapp.site
4. CDVQA (EVALUATION)
Purpose: Change detection VQA
Use: Evaluate bi-temporal analysis
Download: Available on GitHub/HuggingFace
5. For Change Detection Training:
LEVIR-CD: 637 image pairs
WHU-CD: Building change detection
SYSU-CD: Multi-class change detection
Download: All on GitHub/Papers with Code
6. ISRO's OWN DATASET (UNKNOWN)
Cartosat-2S optical images
RISAT SAR images
Co-registered pairs
CANNOT PREPARE SPECIFICALLY
BUT: If your models work on
Sentinel data they'll work on
Cartosat/RISAT too
(similar sensor characteristics)



PART 5: PREPARATION TIMELINE

Month-by-Month Plan:

text

MONTH 1 (July 2026): FOUNDATION

Week 1: Environment Setup

- ☐ Install GDAL, rasterio, pyproj
- ☐ Install PyTorch with CUDA
- ☐ Download BigEarthNet sample (5GB first)
- ☐ Setup GCP with GPU instance
- ☐ Clone GeoChat repository
- ☐ Clone ChangeFormer repository
- ☐ Study BigEarthNet paper thoroughly

Week 2: Data Understanding

- ☐ Load and visualize BigEarthNet
- ☐ Understand Sentinel-1 SAR bands
- ☐ Understand Sentinel-2 MS bands
- ☐ Process first 1000 image pairs
- ☐ Create data loading pipeline
- ☐ Understand GeoTIFF structure

Week 3: Model Exploration

- ☐ Run GeoChat inference on RS images
- ☐ Test ChangeFormer on sample pairs
- ☐ Test RSVG on sample images
- ☐ Evaluate baseline scores on
VRSBench, RSVQA without fine-tuning
- ☐ Document baseline numbers

Week 4: Architecture Design

- ☐ Design complete agent workflow
 - ☐ Design model registry structure
 - ☐ Design input processing pipeline
 - ☐ Create system architecture diagram
 - ☐ Plan fine-tuning strategy
 - ☐ Setup experiment tracking (MLflow/W&B)
-

MONTH 2 (August 2026): CORE DEVELOPMENT

Week 1: Fine-tuning

- ☐ Fine-tune GeoChat on BigEarthNet (Use LoRA for efficiency)
- ☐ Fine-tune on VRSBench training set
- ☐ Fine-tune on RSVQA training set
- ☐ Monitor VQA accuracy metrics
- ☐ Target: >60% accuracy on RSVQA

Week 2: Change Detection

- ☐ Train ChangeFormer on LEVIR-CD
- ☐ Fine-tune on CDVQA training set
- ☐ Build change map visualization
- ☐ Build change description generator
- ☐ Test on CDVQA validation set

Week 3: SAR-Optical Fusion

- ☐ Build SAR feature extractor (CNN trained on BigEarthNet SAR)
- ☐ Build optical feature extractor (ViT on BigEarthNet MS)
- ☐ Build cross-attention fusion layer
- ☐ Test fusion on BigEarthNet pairs
- ☐ Validate on co-registered samples

Week 4: Agentic Layer

- ☐ Build LangGraph orchestrator
 - ☐ Implement model registry
 - ☐ Build query intent classifier
 - ☐ Build input configuration detector
 - ☐ Build task router
 - ☐ Build audit log generator
 - ☐ Test end-to-end pipeline
-

MONTH 3 (September 2026): POLISH + SIH

Week 1: Frontend + Integration

- ☐ Build React web application
- ☐ Image upload + preview component
- ☐ Query input interface
- ☐ Agent status real-time display
- ☐ Results visualization (maps)
- ☐ Execution audit log display
- ☐ PDF report generator

Week 2: Benchmark Evaluation

- ☐ Run full VRSBench evaluation
- ☐ Run full RSVQA evaluation

- ☐ Run full CDVQA evaluation
- ☐ Document all scores
- ☐ Fix weak areas
- ☐ Optimize inference speed

Week 3: SIH HACKATHON

- ☐ 36-hour execution
- ☐ Deploy on GCP
- ☐ Demo preparation
- ☐ Pitch rehearsal

Week 4: After SIH

- ☐ Paper writing (if won!)
- ☐ ISRO follow-up

 PART 6: TECH STACK - COMPLETE
text

COMPLETE TECH STACK FOR SIH26167:

SATELLITE DATA PROCESSING:

- └─ rasterio (GeoTIFF reading)
- └─ GDAL (geospatial operations)
- └─ pyproj (coordinate systems)
- └─ numpy (array operations)
- └─ scipy (signal processing)
- └─ sentinelsat (data download)

COMPUTER VISION + RS MODELS:

- └─ PyTorch (deep learning)
- └─ torchvision (CV operations)
- └─ transformers (HuggingFace)
- └─ GeoChat (RS-VLM)
- └─ RemoteCLIP (RS-CLIP)
- └─ ChangeFormer (change detection)
- └─ RSVG (visual grounding)
- └─ OpenCV (image processing)
- └─ timm (vision models)
- └─ einops (tensor ops)

FINE-TUNING:

- └─ PEFT/LoRA (efficient fine-tuning)
- └─ accelerate (distributed training)
- └─ datasets (HuggingFace datasets)
- └─ evaluate (metrics)
- └─ wandb (experiment tracking)

AGENTIC FRAMEWORK:

- └─ LangGraph (agent orchestration)
- └─ LangChain (tools + chains)
- └─ pydantic (data validation)
- └─ asyncio (async execution)

BACKEND:

- └─ FastAPI (REST API)
- └─ uvicorn (ASGI server)
- └─ websockets (real-time updates)
- └─ celery (task queue)
- └─ redis (task broker)

FRONTEND:

- └─ React (UI framework)
- └─ Leaflet.js (map visualization)
- └─ Chart.js (metrics display)

- └─ Tailwind CSS (styling)
- └─ axios (API calls)

VISUALIZATION:

- └─ matplotlib (change maps)
- └─ folium (geo visualization)
- └─ PIL/Pillow (image manipulation)
- └─ plotly (interactive charts)

DEPLOYMENT:

- └─ Docker (containerization)
- └─ GCP (cloud hosting)
- └─ nginx (reverse proxy)
- └─ GitHub Actions (CI/CD)

 PART 7: WHAT JUDGES WILL EVALUATE
ISRO Evaluation Criteria - Decoded:
text

SCORING BREAKDOWN (Estimated):

1. VRSBench Score (Captioning + VQA + Grounding)

Weight: ~25%

Metric: BLEU, CIDEr, METEOR for captioning
Accuracy for VQA
IoU for grounding

Target scores to win:

VQA Accuracy: >65%

Captioning BLEU-4: >25

Grounding Acc: >60%

2. RSVQA Score

Weight: ~20%

Metric: Overall accuracy

Target: >70% accuracy

(Current SOTA is ~85-90%)

(You need competitive not SOTA)

3. CDVQA Score (Change Detection VQA)

Weight: ~25%

Metric: Change detection F1
Change VQA accuracy

Target: F1 > 0.65 on change maps

VQA Accuracy > 60%

4. ISRO's Cartosat-2S + RISAT Evaluation

Weight: ~20%

Metric: Same as above but on ISRO data

This is the WILDCARD

Cannot prepare specifically

BUT good general models = good here too

5. Agentic Orchestration Quality

Weight: ~10%

Metric: Qualitative assessment
Audit log completeness
Routing correctness

THIS IS WHERE YOU DOMINATE

Your CREVA experience = perfect here

WHAT DISQUALIFIES YOU IMMEDIATELY:

- × Generic VLM without RS adaptation
- × No GeoTIFF support
- × No SAR processing
- × No bi-temporal analysis
- × No audit log
- × Demo that doesn't work on benchmark data



PART 8: YOUR WINNING DIFFERENTIATION

What makes YOUR solution win:
text

STANDARD TEAM APPROACH:

- Use BLIP-2 as is
- Basic change detection
- Simple UI
- Generic agent
- FAIL: "Generic VLM not acceptable"

YOUR WINNING APPROACH:

DIFFERENTIATOR 1: Multi-Model Agentic System

"35 agent systems built at Biencaps"

- = Your agent orchestration will be PRODUCTION QUALITY not student quality
- = Judges will see the difference immediately

DIFFERENTIATOR 2: Proper RS Adaptation

Fine-tune GeoChat on BigEarthNet

- = Not vanilla model = passes ISRO test
- = Benchmark scores will be competitive

DIFFERENTIATOR 3: SAR-Optical Fusion

Most teams will skip this or fake it

- = Complete implementation = extra points
- = ISRO specifically tests this

DIFFERENTIATOR 4: Audit Trail Quality

"Observable execution trace is evaluated"

- = Your LangGraph logs = professional grade
- = CREVA architecture = perfect audit logs

DIFFERENTIATOR 5: UI/UX Quality

Most RS tools look like research code

- = Beautiful React interface = stands out
- = Judges use the interface for 30 minutes

DIFFERENTIATOR 6: Speed

If your inference is fast = better demo

- = GCP GPU optimization = real-time feel
- = Other teams may have slow models



PART 9: RISK ANALYSIS + MITIGATION

text

IDENTIFIED RISKS + SOLUTIONS:

RISK 1: SAR Processing is complex

Probability: HIGH

Impact: HIGH

Mitigation:

- Use BigEarthNet which has SAR included
- Use pretrained SAR encoders
- Focus on feature extraction not raw SAR
- GeoChat already handles some SAR

RISK 2: Fine-tuning takes too long

Probability: MEDIUM

Impact: HIGH

Mitigation:

- Use LoRA (Low Rank Adaptation)
= Fine-tune with 10x less compute
- Use Google Colab Pro+ for GPU
- Start fine-tuning in Week 1
- Use smaller model (7B not 70B)

RISK 3: ISRO dataset is completely different

Probability: LOW-MEDIUM

Impact: HIGH

Mitigation:

- Train on diverse datasets
- Include Indian geographic diversity
- Test on various RS image types
- Strong generalization = handles unknown

RISK 4: Benchmark scores too low

Probability: MEDIUM

Impact: HIGH

Mitigation:

- Start with existing RS models as base
- GeoChat already scores well on RSVQA
- Only need COMPETITIVE not SOTA scores
- Agentic quality compensates for model scores

RISK 5: 36 hours not enough to deploy

Probability: LOW (if prepared)

Impact: VERY HIGH

Mitigation:

- Pre-build entire system before hackathon
- Only PS-specific customization in 36hrs
- Have GCP instance ready to deploy
- Docker containers pre-built
- All models pre-downloaded

RISK 6: Change detection fails on demo

Probability: LOW (if prepared)

Impact: HIGH

Mitigation:

- Test with 20+ image pairs before
- Have backup demo data ready
- Pre-compute results for demo scenarios
- Cache results for reliable demo

 PART 10: 36-HOUR HACKATHON EXECUTION

Assuming full preparation done:

text

HACKATHON DAY PLAN:

HOURL 0-4: DEPLOYMENT

- ☐ Deploy pre-built Docker containers
- ☐ Upload pre-trained models to GCP
- ☐ Verify all APIs responding
- ☐ Test all 5 model endpoints
- ☐ Verify GeoTIFF processing works

- ☐ Load demo images and queries

Hour 4-10: INTEGRATION TESTING

- ☐ Test single image VQA end-to-end
- ☐ Test captioning pipeline
- ☐ Test change detection pipeline
- ☐ Test SAR-optical fusion
- ☐ Test agentic routing
- ☐ Fix any integration bugs

Hour 10-16: UI FINALIZATION

- ☐ Polish React frontend
- ☐ Make map visualization beautiful
- ☐ Add real-time agent status display
- ☐ Make audit log readable
- ☐ Add download report button
- ☐ Mobile responsive check

Hour 16-22: BENCHMARK EVALUATION

- ☐ Run VRSBench test split
- ☐ Run RSVQA test split
- ☐ Run CDVQA test split
- ☐ Document all scores
- ☐ Screenshot results for judges

Hour 22-28: DEMO PREPARATION

- ☐ Prepare 5 killer demo scenarios
- ☐ Pre-load demo images
- ☐ Practice demo flow 3 times
- ☐ Prepare backup video recording
- ☐ Prepare execution screenshots

Hour 28-34: PRESENTATION

- ☐ Build 8-slide presentation
- ☐ Architecture diagram
- ☐ Benchmark scores slide
- ☐ Demo screenshots
- ☐ Impact slide
- ☐ Practice pitch 5 times

Hour 34-36: SLEEP + FINAL CHECK

- ☐ 2 hours mandatory sleep
- ☐ Final system check
- ☐ Backup video confirmed
- ☐ All team members briefed

PART 11: PITCH STRATEGY FOR ISRO JUDGES

ISRO judges are PhD scientists. Pitch accordingly:
text

WHAT ISRO JUDGES CARE ABOUT:

1. Technical rigor (not hype)
2. Benchmark numbers (hard data)
3. Proper RS adaptation (mandatory)
4. SAR handling (their specialty)
5. System architecture quality
6. Real-world deployment potential

WHAT THEY DON'T CARE ABOUT:

- × Fancy buzzwords
- × Generic AI claims
- × UI prettiness alone
- × Vague impact statements

YOUR PITCH STRUCTURE (6 minutes):

[0:00-0:45] TECHNICAL PROBLEM

"Existing RS AI systems are task-specific and require GIS expertise. No system exists that can handle natural language queries across single, bi-temporal, and cross-modal satellite imagery with agentic orchestration"

Speak their language:

- Mention Cartosat-2S and RISAT specifically
- Show you understand SAR limitations
- Reference the cloud penetration advantage
- ISRO judges will nod immediately

[0:45-1:30] YOUR ARCHITECTURE

Show system diagram clearly:

- Input handler with GeoTIFF support
- Agent orchestrator (LangGraph)
- Model registry with 6 specialists
- BigEarthNet fine-tuned base model
- Output layer with visual evidence

Say: "Each component maps directly to your mandatory functional scope requirements"

[1:30-4:00] LIVE DEMO (most important)

Demo Scenario 1 (30 sec):

"Describe land cover in this Sentinel-2 image"

- Show agent selecting captioning model
- Show RS-adapted response (not generic)
- Show execution audit log

Demo Scenario 2 (45 sec):

"What changed between 2020 and 2024?"

- Upload bi-temporal pair
- Show change map generation
- Show change VQA answer

Demo Scenario 3 (45 sec):

"Use optical and SAR to identify built-up and water regions"

- Upload co-registered pair
- Show SAR-optical fusion
- Show combined segmentation map

[4:00-4:45] BENCHMARK SCORES

Show actual numbers:

"On RSVQA: XX% accuracy
On VRSBench VQA: XX%
On CDVQA: XX% F1"

ISRO judges will compare to published state-of-art results immediately
Be honest about gaps but highlight where you're competitive

[4:45-5:30] ISRO-SPECIFIC VALUE

"This system is designed for Cartosat-2S and RISAT compatibility. The same agentic pipeline that works on Sentinel data will work on Indian satellite data with

minimal domain adaptation"

[5:30-6:00] DEPLOYMENT VISION

"Phase 1: NDRF disaster response

Phase 2: Agricultural monitoring

Phase 3: ISRO operational system"

End with confidence not desperation

 PART 12: TEAM COMPOSITION

Who you need for THIS specific PS:
text

TEAM OF 6 - ROLE ASSIGNMENT:

YASH (You) - System Architect + Agent Lead

- LangGraph orchestration
- Model registry design
- Overall system integration
- Pitch delivery (lead presenter)
- This is your domain

MEMBER 2 - RS/CV Engineer (CRITICAL HIRE)

Skills needed:

- Remote sensing background
- GeoTIFF processing experience
- SAR data understanding
- OR willing to learn fast

Role:

- SAR-optical fusion module
- GeoTIFF processing pipeline
- Coordinate system handling
- Rasterio/GDAL operations

MEMBER 3 - ML Engineer (Fine-tuning)

Skills needed:

- PyTorch experience
- HuggingFace familiarity
- Training pipeline knowledge

Role:

- GeoChat fine-tuning on BigEarthNet
- ChangeFormer training
- Benchmark evaluation
- LoRA implementation

MEMBER 4 - Backend Engineer

Skills needed:

- FastAPI
- API design
- Docker

Role:

- FastAPI backend
- Model serving endpoints
- GCP deployment
- Docker containerization

MEMBER 5 - Frontend Engineer

Skills needed:

- React
- Map visualization
- UI/UX

Role:

- React web application
- Leaflet.js maps

- Real-time agent display
- Report PDF generator

MEMBER 6 - Research + QA + Presentation

Skills needed:

- Reading research papers
- Testing mindset
- Good communication

Role:

- Literature review
- Benchmark testing
- Documentation
- Presentation design
- Demo scenario preparation

CRITICAL NOTE:

Member 2 is your MOST IMPORTANT hire
Find someone with even basic remote sensing or GIS exposure
This single person determines if SAR module works or fails

 PART 13: HONEST WIN PROBABILITY
text

FINAL WIN PROBABILITY ASSESSMENT:

IF YOU PREPARE PROPERLY (3 months):

Reach internal college selection:	90%
Reach SIH finale:	65%
Win ANY ISRO prize:	55%
Win consolation (₹15,000):	40%
Win top 3 (₹50,000+):	25%
Win first prize (₹1,00,000):	15%

IF YOU DON'T PREPARE PROPERLY:

Reach internal selection:	60%
Reach SIH finale:	30%
Win ANY prize:	15%

(ISRO judges will reject immediately if mandatory requirements not met)

KEY FACTOR THAT CHANGES EVERYTHING:

Finding Member 2 (RS expert)
With RS expert: +15% all probabilities
Without RS expert: -20% all probabilities

BOTTOM LINE:

This PS is WINNABLE but DEMANDING
It rewards teams that genuinely understand remote sensing
Not teams that fake it with generic AI demos

 IMMEDIATE ACTION LIST
text

STARTING TODAY:

ACTION 1 (Today):

Read this paper completely:
"BigEarthNet: A Large-Scale Benchmark

Archive For Remote Sensing Image Understanding"

- Understand the dataset structure
- Understand SAR + Optical bands

ACTION 2 (Today):

Install and test GeoChat:

- pip install transformers torch
- git clone huggingface/geochat
- Run inference on any satellite image
- See what it outputs currently

ACTION 3 (This Week):

Find your RS-expert team member

- Check your college CS/ECE seniors
- Check for students doing remote sensing mini projects
- Check LinkedIn for RS interns
- This is your most critical task

ACTION 4 (This Week):

Create GCP account + GPU instance

- Get \$300 free credits
- Setup T4 GPU instance
- Test BigEarthNet data loading

ACTION 5 (This Week):

Clone these repositories:

- ☐ github.com/geochat-team/geochat
- ☐ github.com/justchenhao/ChangeFormer
- ☐ github.com/ZhanYang-nwpu/RSVG-pytorch
- ☐ BigEarthNet data loader

ACTION 6 (Submit to SIH):

- ☐ Register team on SIH portal
- ☐ Select PS SIH26167
- ☐ Submit initial idea document
- ☐ Get college SPOC approval

🏆 FINAL STATEMENT text

YASH, HERE IS THE COMPLETE TRUTH:

SIH26167 is the HARDEST PS you could have chosen.

It is also the MOST REWARDING.

ISRO does not give this PS lightly. They want a REAL system. Not a demo. Not a prototype. A WORKING agentic RS AI system.

The teams that will fail:

- Those who use generic VLMs
- Those who skip SAR processing
- Those who don't fine-tune
- Those who build without RS knowledge

The teams that will win:

- Those who read every paper
- Those who fine-tune properly
- Those who build complete pipeline
- Those who score on benchmarks

→ Those whose agents actually work

YOU have one advantage others don't:
Production-level agent architecture
from CREVA + Biencaps internship

USE THAT ADVANTAGE WELL.

The agentic layer is what makes
SatQuery AI novel.
That is YOUR layer.
That is where you WIN.

Everything else can be learned.
Agent architecture is what you ARE.

Start today. 🚀